Assignment 3

`sum = addend + adder`

addendDegree: 3

	7	6	5	4	3	2	1	0
addend					1	-2	0	-4

$x^3 - 2x^2 - 4$

adderDegree: 1

	7	6	5	4	3	2	1	0
adder							1	-3

$x - 3$

sumDegree:

	7	6	5	4	3	2	1	0
sum								

`sum = addend + adder`

addendDegree $\boxed{3}$

	7	6	5	4	3	2	1	0
addend					1	-2	0	-4

$$x^3 - 2x^2 - 4$$

adderDegree $\boxed{1}$

	7	6	5	4	3	2	1	0
adder							1	-3

$$x - 3$$

sumDegree $\boxed{3}$

	7	6	5	4	3	2	1	0
sum					1	-2	1	-7

$$x^3 - 2x^2 + x - 7$$

`sum = addend + adder`

addendDegree: 1

	7	6	5	4	3	2	1	0
addend							1	-3

$x - 3$

adderDegree: 3

	7	6	5	4	3	2	1	0
adder					1	-2	0	-4

$x^3 - 2x^2 - 4$

sumDegree:

	7	6	5	4	3	2	1	0
sum								

sum = addend + adder

addendDegree: 1

	7	6	5	4	3	2	1	0	
addend							1	-3	$x - 3$

adderDegree: 3

	7	6	5	4	3	2	1	0	
adder					1	-2	0	-4	$x^3 - 2x^2 - 4$

sumDegree: 3

	7	6	5	4	3	2	1	0	
sum					1	-2	1	-7	$x^3 - 2x^2 + x - 7$

`sum = addend + adder`

addendDegree: 3

	7	6	5	4	3	2	1	0
addend					1	-2	0	-4

$x^3 - 2x^2 - 4$

adderDegree: 0

	7	6	5	4	3	2	1	0
adder								0

0

sumDegree:

	7	6	5	4	3	2	1	0
sum								

`sum = addend + adder`

addendDegree: 3

	7	6	5	4	3	2	1	0
addend					1	-2	0	-4

$x^3 - 2x^2 - 4$

adderDegree: 0

	7	6	5	4	3	2	1	0
adder								0

0

sumDegree: 3

	7	6	5	4	3	2	1	0
sum					1	-2	0	-4

$x^3 - 2x^2 - 4$

`sum = addend + adder`

			7	6	5	4	3	2	1	0	
addendDegree	0	addend								0	0

			7	6	5	4	3	2	1	0	
adderDegree	3	adder					1	-2	0	-4	$x^3 - 2x^2 - 4$

			7	6	5	4	3	2	1	0
sumDegree		sum								

`sum = addend + adder`

addendDegree $\boxed{0}$

	7	6	5	4	3	2	1	0	
addend								0	0

adderDegree $\boxed{3}$

	7	6	5	4	3	2	1	0	
adder					1	-2	0	-4	$x^3 - 2x^2 - 4$

sumDegree $\boxed{3}$

	7	6	5	4	3	2	1	0	
sum					1	-2	0	-4	$x^3 - 2x^2 - 4$

sum = addend + adder

addendDegree: 3

addend	7	6	5	4	3	2	1	0
					-1	-2	0	4

$-x^3 - 2x^2 + 4$

adderDegree: 3

adder	7	6	5	4	3	2	1	0
					1	2	1	-7

$x^3 + 2x^2 + x - 7$

sumDegree:

sum	7	6	5	4	3	2	1	0

`sum = addend + adder`

`addendDegree` 3

`adderDegree` 3

`sumDegree` 1

	7	6	5	4	3	2	1	0	
`addend`					-1	-2	0	4	$-x^3 - 2x^2 + 4$
`adder`					1	2	1	-7	$x^3 + 2x^2 + x - 7$
`sum`					0	0	1	-3	$x - 3$

`product = multiplicand * multiplier`

multiplicandDegree $\boxed{3}$

multiplicand

4	3	2	1	0
	1	-2	0	-4

$x^3 - 2x^2 - 4$

multiplierDegree $\boxed{1}$

multiplier

4	3	2	1	0
			1	-3

$x - 3$

productDegree $\boxed{}$

product

4	3	2	1	0

```
product = multiplicand * multiplier
```

			4	3	2	1	0	
multiplicandDegree	3	multiplicand		1	-2	0	-4	$x^3 - 2x^2 - 4$

			4	3	2	1	0	
multiplierDegree	1	multiplier				1	-3	$x - 3$

			4	3	2	1	0	
productDegree	3	product		-3	6	0	12	$-x^3 + 6x^2 + 12$

```
product = multiplicand * multiplier
```

			4	3	2	1	0	
multiplicandDegree	3	multiplicand		1	-2	0	-4	$x^3 - 2x^2 - 4$
multiplierDegree	1	multiplier				1	-3	$x - 3$
productDegree	4	product	1	-5	6	-4	12	

$$-x^4 - 5x^3 + 6x^2 - 4x + 12$$

`product = multiplicand * multiplier`

`multiplicandDegree` = 3

`multiplicand`

4	3	2	1	0
	1	-2	0	-4

$x^3 - 2x^2 - 4$

`multiplierDegree` = 0

`multiplier`

4	3	2	1	0
				0

0

`productDegree`

`product`

4	3	2	1	0

`product = multiplicand * multiplier`

multiplicandDegree: 3

multiplicand	4	3	2	1	0
		1	-2	0	-4

$x^3 - 2x^2 - 4$

multiplierDegree: 0

multiplier	4	3	2	1	0
					0

0

productDegree: 0

product	4	3	2	1	0
					0

0

`product = multiplicand * multiplier`

multiplicandDegree	multiplicand	4	3	2	1	0	
0						0	0

multiplierDegree	multiplier	4	3	2	1	0	
1					1	-3	$x - 3$

productDegree	product	4	3	2	1	0

```
product = multiplicand * multiplier
```

			4	3	2	1	0	
multiplicandDegree	0	multiplicand					0	0

			4	3	2	1	0	
multiplierDegree	1	multiplier				1	-3	$x - 3$

			4	3	2	1	0	
productDegree	0	product					0	0

```
quotient = dividend / divisor;
remainder = dividend % divisor
```

$$x - 3 \,\overline{)\, x^3 - 2x^2 + 0x - 4}$$

remainderDegree $\boxed{3}$

	3	2	1	0
remainder	1	-2	0	-4

divisorDegree $\boxed{1}$

	3	2	1	0
divisor			1	-3

bufferDegree $\boxed{3}$

	3	2	1	0
buffer	0	0	0	0

quotientDegree $\boxed{2}$

	3	2	1	0
quotient		0	0	0

i $\boxed{2}$

$$\begin{array}{r|l} & x^2 \\ \hline x - 3 & x^3 - 2x^2 + 0x - 4 \end{array}$$

remainderDegree: $\boxed{3}$

remainder

3	2	1	0
1	-2	0	-4

divisorDegree: $\boxed{1}$

divisor

3	2	1	0
		1	-3

bufferDegree: $\boxed{3}$

buffer

3	2	1	0
0	0	0	0

quotientDegree: $\boxed{2}$

quotient

3	2	1	0
	1	0	0

i: $\boxed{2}$

$$
\begin{array}{r|l}
 & x^2 \\
\hline
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & x^3 - 3x^2
\end{array}
$$

remainderDegree $\boxed{3}$

	3	2	1	0
remainder	1	-2	0	-4

divisorDegree $\boxed{1}$

	3	2	1	0
divisor			1	-3

bufferDegree $\boxed{3}$

	3	2	1	0
buffer	1	-3	0	0

quotientDegree $\boxed{2}$

	3	2	1	0
quotient		1	0	0

i $\boxed{2}$

$$
\begin{array}{r|l}
 & x^2 \\ \hline
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & x^2 + 0x - 4
\end{array}
$$

remainderDegree	2

	3	2	1	0
remainder		1	0	0

divisorDegree	1

	3	2	1	0
divisor			1	-3

bufferDegree	3

	3	2	1	0
buffer	1	-3	0	0

quotientDegree	2

	3	2	1	0
quotient		1	0	0

i	2

$$
\begin{array}{r|l}
 & x^2 \\
\hline
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & x^3 - 3x^2 \\
\cline{2-2}
 & \qquad\quad x^2 + 0x - 4
\end{array}
$$

remainderDegree: 2

	3	2	1	0
remainder		1	0	0

divisorDegree: 1

	3	2	1	0
divisor			1	-3

bufferDegree: 3

	3	2	1	0
buffer	1	-3	0	0

quotientDegree: 2

	3	2	1	0
quotient		1	0	0

i: 1

$$
\begin{array}{r|l}
 & x^2 + x \\
\cline{2-2}
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & \qquad\quad x^2 + 0x - 4
\end{array}
$$

remainderDegree: 2

remainder	3	2	1	0
		1	0	0

divisorDegree: 1

divisor	3	2	1	0
			1	-3

bufferDegree: 3

buffer	3	2	1	0
	1	-3	0	0

quotientDegree: 2

quotient	3	2	1	0
		1	1	0

i: 1

$$
\begin{array}{r|l}
 & x^2 + x \\
\hline
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & \qquad\quad x^2 + 0x - 4 \\
 & \qquad\quad x^2 - 3x
\end{array}
$$

remainderDegree: 2

	3	2	1	0
remainder		1	0	0

divisorDegree: 1

	3	2	1	0
divisor			1	-3

bufferDegree: 2

	3	2	1	0
buffer		1	-3	0

quotientDegree: 2

	3	2	1	0
quotient		1	1	0

i: 1

$$
\begin{array}{r|l}
 & x^2 + x \\
\cline{2-2}
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & x^3 - 3x^2 \\
\cline{2-2}
 & \qquad x^2 + 0x - 4 \\
 & \qquad x^2 - 3x \\
\cline{2-2}
 & \qquad\qquad 3x - 4
\end{array}
$$

remainderDegree $\boxed{1}$

remainder

3	2	1	0
		3	-4

divisorDegree $\boxed{1}$

divisor

3	2	1	0
		1	-3

bufferDegree $\boxed{2}$

buffer

3	2	1	0
	1	-3	0

quotientDegree $\boxed{2}$

quotient

3	2	1	0
	1	1	0

i $\boxed{1}$

$$
\begin{array}{r l}
 & x^2 + x \\
x - 3 \,\big) & \overline{x^3 - 2x^2 + 0x - 4} \\
 & \underline{x^3 - 3x^2} \\
 & \qquad\quad x^2 + 0x - 4 \\
 & \qquad\quad \underline{x^2 - 3x} \\
 & \qquad\qquad\quad 3x - 4
\end{array}
$$

			3	2	1	0
remainderDegree	1	remainder			3	-4

			3	2	1	0
divisorDegree	1	divisor			1	-3

			3	2	1	0
bufferDegree	2	buffer		1	-3	0

			3	2	1	0		
quotientDegree	2	quotient		1	1	0	i	0

$$
\begin{array}{r|l}
 & x^2 + x + 3 \\
\hline
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & \quad\; x^2 + 0x - 4 \\
 & \quad\; \underline{x^2 - 3x} \\
 & \qquad\quad 3x - 4
\end{array}
$$

			3	2	1	0
remainderDegree	1	remainder			3	-4

			3	2	1	0
divisorDegree	1	divisor			1	-3

			3	2	1	0
bufferDegree	2	buffer		1	-3	0

			3	2	1	0		
quotientDegree	2	quotient		1	1	3	i	0

$$
\begin{array}{r|l}
 & x^2 + x + 3 \\
\cline{2-2}
x - 3 & x^3 - 2x^2 + 0x - 4 \\
 & x^3 - 3x^2 \\
\cline{2-2}
 & x^2 + 0x - 4 \\
 & x^2 - 3x \\
\cline{2-2}
 & 3x - 4 \\
 & 3x - 9
\end{array}
$$

remainderDegree	remainder	3	2	1	0
1				3	-4

divisorDegree	divisor	3	2	1	0
1				1	-3

bufferDegree	buffer	3	2	1	0
1				3	-9

quotientDegree	quotient	3	2	1	0
2			1	1	3

i: 0

$$
\begin{array}{r}
x^2 + x + 3 \\
x - 3 \overline{\big)\, x^3 - 2x^2 + 0x - 4} \\
\underline{x^3 - 3x^2 } \\
x^2 + 0x - 4 \\
\underline{x^2 - 3x } \\
3x - 4 \\
\underline{3x - 9} \\
5
\end{array}
$$

remainderDegree $\boxed{0}$

remainder	3	2	1	0
				5

divisorDegree $\boxed{1}$

divisor	3	2	1	0
			1	-3

bufferDegree $\boxed{1}$

buffer	3	2	1	0
			3	-9

quotientDegree $\boxed{2}$

quotient	3	2	1	0
		1	1	3

i $\boxed{0}$

$$x - 3 \,\overline{)\, x^3 - 2x^2 - 3x - 4}$$

			3	2	1	0
remainderDegree	3	remainder	1	-2	-3	-4

			3	2	1	0
divisorDegree	1	divisor			1	-3

			3	2	1	0
bufferDegree	3	buffer	0	0	0	0

			3	2	1	0		
quotientDegree	2	quotient		0	0	0	i	2

$$\begin{array}{r|l} & x^2 \\ \hline x - 3 & x^3 - 2x^2 - 3x - 4 \end{array}$$

remainderDegree $\boxed{3}$

remainder

3	2	1	0
1	-2	-3	-4

divisorDegree $\boxed{1}$

divisor

3	2	1	0
		1	-3

bufferDegree $\boxed{3}$

buffer

3	2	1	0
0	0	0	0

quotientDegree $\boxed{2}$

quotient

3	2	1	0
	1	0	0

i $\boxed{2}$

$$
\begin{array}{r l}
 & x^2 \\
x - 3 \enclose{longdiv}{} & \overline{x^3 - 2x^2 - 3x - 4} \\
 & x^3 - 3x^2
\end{array}
$$

remainderDegree: 3

	3	2	1	0
remainder	1	-2	-3	-4

divisorDegree: 1

	3	2	1	0
divisor			1	-3

bufferDegree: 3

	3	2	1	0
buffer	1	-3	0	0

quotientDegree: 2

	3	2	1	0
quotient		1	0	0

i: 2

$$
\begin{array}{r|l}
 & x^2 \\ \cline{2-2}
x - 3 & x^3 - 2x^2 - 3x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & \qquad\;\; x^2 - 3x - 4
\end{array}
$$

			3	2	1	0
remainderDegree	2	remainder		1	-3	0

			3	2	1	0
divisorDegree	1	divisor			1	-3

			3	2	1	0
bufferDegree	3	buffer	1	-3	0	0

			3	2	1	0		
quotientDegree	2	quotient		1	0	0	i	2

$$
\begin{array}{r|l}
 & x^2 \\
\cline{2-2}
x - 3 & x^3 - 2x^2 - 3x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & \qquad\;\; x^2 - 3x - 4
\end{array}
$$

			3	2	1	0
remainderDegree	2	remainder		1	-3	0

			3	2	1	0
divisorDegree	1	divisor			1	-3

			3	2	1	0
bufferDegree	3	buffer	1	-3	0	0

			3	2	1	0		
quotientDegree	2	quotient		1	0	0	i	1

$$
\begin{array}{rl}
 & x^2 + x \\
x - 3 \,\big) & x^3 - 2x^2 - 3x - 4 \\
 & x^3 - 3x^2 \\
\hline
 & \quad\quad x^2 - 3x - 4
\end{array}
$$

			3	2	1	0
remainderDegree	2	remainder		1	-3	0

			3	2	1	0
divisorDegree	1	divisor			1	-3

			3	2	1	0
bufferDegree	3	buffer	1	-3	0	0

			3	2	1	0		
quotientDegree	2	quotient		1	1	0	i	1

$$
\begin{array}{r|l}
 & x^2 + x \\
\hline
x - 3 & x^3 - 2x^2 - 3x - 4 \\
 & \underline{x^3 - 3x^2} \\
 & \qquad\; x^2 - 3x - 4 \\
 & \qquad\; x^2 - 3x
\end{array}
$$

remainderDegree $\boxed{2}$

remainder	3	2	1	0
		1	-3	0

divisorDegree $\boxed{1}$

divisor	3	2	1	0
			1	-3

bufferDegree $\boxed{2}$

buffer	3	2	1	0
		1	-3	0

quotientDegree $\boxed{2}$

quotient	3	2	1	0
		1	1	0

i $\boxed{1}$

$$
\begin{array}{r|l}
 & x^2 + x \\
\hline
x - 3 & x^3 - 2x^2 - 3x - 4 \\
 & x^3 - 3x^2 \\
\cline{2-2}
 & x^2 - 3x - 4 \\
 & x^2 - 3x \\
\cline{2-2}
 & - 4
\end{array}
$$

remainderDegree $\boxed{0}$

remainder	3	2	1	0
				-4

divisorDegree $\boxed{1}$

divisor	3	2	1	0
			1	-3

bufferDegree $\boxed{2}$

buffer	3	2	1	0
		1	-3	0

quotientDegree $\boxed{2}$

quotient	3	2	1	0
		1	1	0

i $\boxed{1}$